

Indian Knowledge Systems (IKS): Foundations, Philosophy, and Contributions

1. Overview and Core Philosophy

Indian Knowledge Systems (IKS) encompass the vast corpus of traditional knowledge, philosophy, scientific methodologies, mathematics, metallurgy, architecture, astronomy, and health sciences developed on the Indian subcontinent over millennia. At the core of IKS is an integrated worldview that perceives knowledge as holistic, connecting humanity, nature, and the universe.

2. Primary Textual Corpus and Vedic Foundations

The foundation of IKS resides in the Vedic literature: Rigveda, Samaveda, Yajurveda, and Atharvaveda, accompanied by the Aranyakas, Brahmanas, and Upanishads. Complementing these are the six Vedangas: Shiksha (phonetics), Kalpa (rituals and social law), Vyakarana (grammar), Nirukta (etymology), Chandas (prosody), and Jyotisha (astronomy). Panini's Ashtadhyayi stands as a monumental work in formal linguistics and generative grammar.

3. Mathematics and Astronomy

Ancient and medieval Indian mathematicians made pioneering contributions to global science. Aryabhata introduced foundational concepts of algebra, spherical trigonometry, and proposed that the Earth rotates on its axis. Brahmagupta established the mathematical principles of zero (Shunya) as a number and rules for negative integers. Bhaskaracharya elaborated on calculus principles, planetary positions, and the cyclic method (Chakravala) for solving indeterminate quadratic equations.

4. Health Sciences: Ayurveda and Yoga

Ayurveda, detailed in classic treatises like Charaka Samhita and Sushruta Samhita, emphasizes preventive health, balancing the three doshas (Vata, Pitta, Kapha), and holistic well-being. Sushruta is celebrated for pioneering surgical techniques, including rhinoplasty and cataract surgeries. Patanjali's Yoga Sutras provide a systematic eight-limbed framework (Ashtanga Yoga) for mental discipline, focus, and spiritual development.

5. Metallurgy, Architecture, and Sustainable Practices

Historical Indian metallurgy achieved extraordinary feats such as the rust-resistant Iron Pillar of Delhi and the casting of high-grade Wootz steel. Vastu Shastra and temple architecture reflected sophisticated geometric planning, acoustic engineering, and environmental harmony. Agricultural and water management practices, such as stepwells (Baolis) and Ahar-Pyne systems, demonstrated sustainable ecological stewardship.